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GEOGRAPHY · Andhra Pradesh: physiography, climate and drainage

UNDERSTAND IT FIRST

What is it? Physiography means the **shape of the land**: plains, hills, plateaus. For Andhra Pradesh it explains **why the deltas are rice bowls, why Rayalaseema faces drought, and why the coast is hit by cyclones**. AP lies on the **east coast** between **12°N and 19°N**, bordered by **Odisha, Chhattisgarh, Telangana, Karnataka and Tamil Nadu**, and the **Bay of Bengal**. Learn it as **location and size → three physical divisions → hills, caves and lakes → seasons and monsoons → cyclones → rivers north to south → irrigation projects**.

1. Location and physical divisions

- **Size:** about **1.63 lakh sq km** (about **5%** of India; **7th** largest State by area). **Coastline:** officially about **974 km**, the **second longest** after Gujarat (a newer survey measured about **1,053 km**; check the figure your source uses). **Districts: 26** from **April 2022**; **Polavaram** and **Markapuram** districts were added on **31 December 2025**, making **28** (check for later changes). Capital: **Amaravati**.

Physical divisions of Andhra Pradesh

Division	Features
Coastal plains	A strip from **Srikakulam to Tirupati district**, widest in the **Godavari–Krishna deltas** (rich alluvium; the **'rice bowl'**). **Kolleru** (freshwater lake between the two deltas; **Ramsar site**, pelicans), **Pulicat** (brackish lagoon, second largest after Chilika; **Sriharikota** island separates it from the sea; flamingos at **Nelapattu–Pulicat**). Sand ridges, backwaters and mangroves (**Coringa**, Krishna mangroves).
Eastern Ghats	Not a continuous range; **broken by the Godavari, Krishna and Pennar**. **North:** the hills of **Alluri Sitharama Raju, Parvathipuram Manyam** districts: **Araku valley, Paderu, Anantagiri**, **Arma Konda (about 1,680 m)**, usually given as the **highest peak in AP**; the **Papikondalu** gorge on the Godavari. **South: Nallamala** (largest; Srisailam, **Nagarjunasagar–Srisailam Tiger Reserve**, India's largest), **Velikonda, Lankamala, Palakonda, Seshachalam** (Tirumala; red sanders), **Horsleykonda** (hill station, Annamayya district).
Rayalaseema plateau	Western part of AP, a section of the **Deccan plateau** made of old **granite and gneiss**; **red soils**, black soils in parts; low, uncertain rain. **Gandikota gorge** on the Pennar (YSR Kadapa), the **'Grand Canyon of India'**; **Belum caves** (Nandyal), among the **longest caves on the Indian plains**; tanks and borewells are the main water sources. In the north, **Borra caves** (limestone, **Ananthagiri hills**, on the **Gosthani** river) are among India's largest caves.

2. Climate

Seasons and rainfall in Andhra Pradesh

- AP has a **tropical monsoon** climate: hot summers, mild winters. Normal annual rain is roughly **950 mm**. The **south-west monsoon (June–September)** brings **most of the rain** to the State, especially the **north coast and Agency hills**. The **north-east (retreating) monsoon (October–December)** is the **main rainy season for south coastal AP** (Nellore, Prakasam, Tirupati) and southern Rayalaseema, because winds from the Bay of Bengal reach them directly.
- **Anantapur** is the **driest** district (about **550 mm**; among the lowest in India after western Rajasthan) because it lies in the **rain-shadow** of the Western Ghats and is far from the Bay; it is **drought-prone**. Summer temperatures cross **45°C** in **Rayalaseema, Palnadu and Prakasam**; **heatwaves** kill people most years. **Cyclones** form in the Bay mainly in **October–December** (also May): the **Diviseema cyclone (1977)** with a huge storm surge, **Hudhud (2014, Visakhapatnam)**, **Titli (2018, Srikakulam)**, **Michaung (2023, near Bapatla)** and **Montha (October 2025, near Kakinada)**. The **AP State Disaster Management Authority** issues warnings; cyclone shelters and mangroves reduce damage.

3. Rivers and projects

Main rivers of Andhra Pradesh

River	In Andhra Pradesh
Godavari	Enters AP below Bhadrachalam; the **Sabari** joins it; cuts the **Papikondalu** gorge; **Polavaram (Indira Sagar) project**, a **national project**; at **Dowleswaram** (near Rajamahendravaram) **Sir Arthur Cotton's anicut (1852)**, now a barrage, irrigates the delta; splits into the **Gautami** and **Vasishta** branches; meets the sea near **Antarvedi** and **Yanam**.
Krishna	The **Tungabhadra** joins at **Sangameswaram** (Nandyal); **Srisailam** dam (power; feeds the **Handri–Neeva**, **Telugu Ganga** and **Galeru–Nagari** canals for Rayalaseema), **Nagarjuna Sagar** (one of the largest masonry dams), **Pulichintala**, **Prakasam barrage (Vijayawada)**; delta in Krishna and Bapatla districts; mouth near **Hamsaladeevi**. **Pattiseema (2016)** and **Purushottapatnam** lift schemes move **Godavari water into the Krishna**.
Pennar (Penna)	Rises in the **Nandi hills** (Karnataka); flows through **Anantapur, YSR Kadapa and Nellore**; tributaries **Chitravati, Papagni, Kunderu (Kundu), Sagileru, Cheyyeru**; **Gandikota** gorge and reservoir, **Mylavaram**, **Somasila** dam (Nellore); reaches the sea near Nellore. The **Telugu Ganga project (1983)** also carries Krishna water to **Chennai** for drinking.
Smaller rivers	**North coast: Vamsadhara** (from Odisha; **Gotta barrage**; mouth at **Kalingapatnam**; AP–Odisha tribunal **2017**), **Nagavali (Langulya)** (from Odisha; **Thotapalli** project), **Champavati, Gosthani, Sarada, Varaha, Tandava, Yeleru**. **Between the deltas and Nellore: Gundlakamma** (Nallamala; **Gundlakamma reservoir**), **Musi, Paleru, Manneru**. **South: Swarnamukhi** (past **Srikalahasti**), **Kalangi**, **Araniar**. **Inland lakes:** Kolleru; tanks are vital in Rayalaseema.

EXAM TRAP

AP **12°–19°N**; **~1.63 lakh km², 7th**; coast **~974 km (2nd after Gujarat; new survey ~1,053)**; **26 districts (2022) → 28 (31 Dec 2025: Polavaram, Markapuram)**; divisions: **coastal plains** (deltas, **Kolleru Ramsar**, **Pulicat** 2nd lagoon, **Sriharikota**, Coringa), **Eastern Ghats** (north: **Araku, Paderu, Anantagiri, Arma Konda ~1,680 m, Papikondalu**; south: **Nallamala (NSTR largest tiger reserve), Velikonda, Lankamala, Palakonda, Seshachalam, Horsleykonda**), **Rayalaseema plateau** (granite, red soils; **Gandikota 'Grand Canyon'**, **Belum caves**); **Borra caves (Gosthani, limestone)**; climate: rain **~950 mm**; **SW monsoon most rain**; **NE monsoon main for south coast**; **Anantapur driest (~550 mm, rain-shadow)**; **45°C+ heatwaves**; cyclones **Oct–Dec**: **Diviseema 1977, Hudhud 2014, Titli 2018, Michaung 2023, Montha Oct 2025**; **Godavari**: Sabari, Papikondalu, **Polavaram (national)**, **Dowleswaram (Cotton, 1852)**, **Gautami, Vasishta**, Antarvedi; **Krishna**: **Tungabhadra at Sangameswaram**, **Srisailam, Nagarjuna Sagar, Pulichintala, Prakasam barrage**, **Hamsaladeevi**; **Pattiseema 2016, Purushottapatnam**; **Handri–Neeva, Galeru–Nagari, Telugu Ganga (1983, Chennai)**; **Pennar**: Nandi hills; Chitravati, Papagni, Kunderu; **Gandikota, Somasila**; north: **Vamsadhara (Gotta; Kalingapatnam; 2017 award), Nagavali (Thotapalli)**, Gosthani, Sarada; **Gundlakamma**; **Swarnamukhi (Srikalahasti)**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Andhra Pradesh has the second longest coastline among Indian States.

2. The south-west monsoon brings most of Andhra Pradesh's annual rainfall.

3. Anantapur is one of the driest districts of the State.

4. The Krishna splits into the Gautami and Vasishta branches at Dowleswaram.

Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. It is the Godavari, not the Krishna, that splits into the Gautami and Vasishta at Dowleswaram.

PRACTICE

Match List-I (feature) with List-II (district):

List-I (feature)	List-II (district)
A. Belum caves	**1.** YSR Kadapa
B. Borra caves	**2.** Tirupati

C. Gandikota gorge	**3.** Nandyal
D. Pulicat lake	**4.** Alluri Sitharama Raju

(1) A → 4 B → 3 C → 1 D → 2

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 3 B → 4 C → 1 D → 2

(4) A → 1 B → 4 C → 3 D → 2

Answer: (3) Belum: Nandyal; Borra: Alluri Sitharama Raju; Gandikota: YSR Kadapa; Pulicat: Tirupati.

PRACTICE

Which one of the following statements is NOT correct?

A. The Tungabhadra joins the Krishna at Sangameswaram.

B. Kolleru lake lies between the Godavari and Krishna deltas.

C. Sriharikota island separates Pulicat lake from the sea.

D. The north-east monsoon is the main rainy season for north coastal Andhra.

Answer: D. The north-east monsoon is the main rainy season for south coastal AP and southern Rayalaseema; the north coast gets most rain from the south-west monsoon.

PRACTICE

Assertion (A): Much of Rayalaseema, especially Anantapur, receives low and uncertain rainfall.
Reason (R): The region lies in the rain-shadow of the Western Ghats and is far from the Bay of Bengal.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Both are true, and the rain-shadow position and distance from the Bay explain the low rainfall.

PRACTICE

A river rises in Odisha, flows through Srikakulam district and meets the sea at Kalingapatnam; its water dispute was settled by a tribunal in 2017. Which river is it?

A. Nagavali

B. Vamsadhara

C. Sarada

D. Gosthani

Answer: B. The Vamsadhara meets the sea at Kalingapatnam; the Vamsadhara Water Disputes Tribunal gave its award in 2017.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write AP's location, size, coastline, districts and three physical divisions with hills, caves, lakes and islands. 2. Write the seasons, the two monsoons by region, dry and wet areas, and the cyclones with years. 3. Write the rivers north to south with tributaries, dams, barrages and projects.

ECONOMY · Indian banking: structure, development and reforms

UNDERSTAND IT FIRST

What is it? A **bank** accepts **deposits** from people who save and gives **loans** to people who need money, earning the **difference in interest**. Banks carry savings to farms, factories and homes, so a sound banking system is the **backbone of growth**. India's banks grew from a few colonial banks to a vast network; **nationalisation** took them to villages, and the **1991 reforms** made them efficient and competitive. Learn it as **structure → history up to nationalisation → rural banking → reforms after 1991 → recent changes**.

1. Structure

Structure of Indian banking

- **Scheduled banks** are listed in the **Second Schedule of the RBI Act, 1934** (paid-up capital and reserves of at least **₹5 lakh**, and sound affairs); they can borrow from the RBI and must keep CRR with it. **Commercial banks**: **public sector** (**12** since the **April 2020 mergers**: SBI, PNB, Bank of Baroda, Canara, Union, Indian Bank, Bank of India, Central Bank, Indian Overseas, UCO, Bank of Maharashtra, Punjab & Sind), **private**

(HDFC Bank, ICICI, Axis and others), **foreign** banks, and **RRBs**. **Differentiated banks** (on the **Nachiket Mor Committee, 2013**; licences **2015**): **small finance banks** (small loans to farmers, small units; **75%** priority sector) and **payments banks** (accept deposits **up to ₹2 lakh** per customer, **cannot lend**; e.g. India Post Payments Bank).

- **Regional Rural Banks (RRBs):** set up on the **Narasimham Working Group (1975)**, first on **2 October 1975** (**Prathama Bank**, Moradabad), under the **RRB Act 1976**; owned **Centre 50%, sponsor bank 35%, State 15%**; serve small farmers, labourers and artisans. Under **'One State, One RRB'**, they were amalgamated into **28 RRBs from 1 May 2025**; AP's four RRBs became one **Andhra Pradesh Grameena Bank**. **Co-operative banks**: **urban co-operative banks**, and a rural **three-tier** structure: **State Co-operative Bank → District Central Co-operative Banks → Primary Agricultural Credit Societies (PACS)**; regulated by **both the RBI and State Registrars**; the **Banking Regulation (Amendment) Act 2020** brought them more firmly under the RBI.

2. Development up to nationalisation

Milestones of Indian banking

- **Early banks: Bank of Hindustan (1770)**, the first; **General Bank of India (1786)**. The three **Presidency Banks**, **Bank of Bengal (1806, as Bank of Calcutta)**, **Bank of Bombay (1840)** and **Bank of Madras (1843)**, merged into the **Imperial Bank of India (1921)**. **Allahabad Bank (1865)**; **Punjab National Bank (1894)**, the first bank run wholly with **Indian capital**, linked with **Lala Lajpat Rai**; the **Swadeshi movement** inspired more Indian banks. **Andhra Bank** was founded in **1923 at Machilipatnam** by **Dr Bhogaraju Pattabhi Sitaramayya** (merged into **Union Bank** in 2020).
- **After Independence:** the **Banking Regulation Act, 1949** gave the RBI wide powers. On the **All India Rural Credit Survey (Gorwala Committee, 1954)**, the Imperial Bank became the **State Bank of India (1 July 1955)**; its associate banks were merged into SBI in **2017**. **Social control (1967)**, then **nationalisation**: **14 banks** with deposits over **₹50 crore** on **19 July 1969**, and **6 more** with deposits over **₹200 crore** on **15 April 1980**. Aims: take banks to **villages**, end **control by big business**, lend to **agriculture and small industry** (priority sector). The **Lead Bank Scheme (1969)** (Gadgil Study Group, Nariman Committee) gave each district a **lead bank**. **NABARD (12 July 1982)** became the apex bank for rural credit.

3. Reforms after 1991

Reform	What it did
Narasimham Committee I (1991)	Cut **SLR and CRR** step by step; **freed interest rates**; **prudential norms** for income recognition and **NPAs** (loans unpaid for **90 days** became NPAs from 2004); **capital adequacy** under the **Basel** norms; entry of **new private banks (1993)**: UTI Bank (now Axis), ICICI, HDFC Bank and others.
Narasimham Committee II (1998)	Merge strong banks; stronger capital; deal with bad loans through **Asset Reconstruction Companies**; less government control.

Recovering bad loans	**Debt Recovery Tribunals (1993)**; **SARFAESI Act 2002** (banks can seize and sell secured assets without going to court); **Insolvency and Bankruptcy Code 2016** (time-bound resolution); **NARCL ('bad bank', 2021)**; the **'4R' approach** (recognition, resolution, recapitalisation, reform) after the **2015 Asset Quality Review**. Gross NPAs fell from over **11% (2018)** to about **2–3%** by 2025 (check the latest).
Governance, mergers, inclusion	**P. J. Nayak Committee (2014)** on bank boards → **Indradhanush (2015)** and the **Banks Board Bureau (2016)**, now the **FSIB (2022)**; mergers: Bank of Baroda + Vijaya + Dena (**2019**), then **10 banks into 4 (April 2020)**, leaving **12 PSBs**; **PM Jan Dhan Yojana (August 2014)**; **DICGC** deposit cover raised to **₹5 lakh (2020)**; **RBI Integrated Ombudsman (2021)**; the **Banking Laws (Amendment) Act, 2025** allows **up to four nominees** per account.

EXAM TRAP

Scheduled = 2nd Schedule RBI Act (₹5 lakh); commercial: **12 PSBs (April 2020)**, private, foreign; **Nachiket Mor 2013 → SFBs (75% PSL), payments banks (₹2 lakh, no loans) 2015**; **RRBs: Narasimham WG 1975, Prathama Bank 2 Oct 1975, Act 1976, 50:35:15; 28 RRBs from 1 May 2025; AP Grameena Bank**; co-op **3-tier: StCB → DCCB → PACS**; **BR (Amendment) Act 2020**; **Bank of Hindustan 1770**; **Presidency banks 1806/1840/1843 → Imperial 1921 → SBI 1 July 1955 (Gorwala, AIRCS 1954)**; **PNB 1894 (Lala Lajpat Rai)**; **Andhra Bank 1923 Machilipatnam (Pattabhi Sitaramayya; Union Bank 2020)**; **BR Act 1949**; **nationalisation 14 (19 July 1969, >₹50 cr) + 6 (15 April 1980, >₹200 cr)**; **Lead Bank 1969**; **NABARD 1982**; **Narasimham I 1991** (SLR/CRR cut, free rates, NPA norms, Basel, **private banks 1993**), **II 1998**; **DRT 1993, SARFAESI 2002, IBC 2016, NARCL 2021, 4R, AQR 2015**; **P. J. Nayak 2014, Indradhanush 2015, BBB 2016 → FSIB 2022**; **mergers 2019, 2020**; **Jan Dhan Aug 2014**; **DICGC ₹5 lakh (2020)**; **Ombudsman 2021**; **2025 Act: 4 nominees**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Fourteen major banks were nationalised on 19 July 1969.

2. The State Bank of India was formed in 1955 from the Imperial Bank of India.

3. Andhra Bank was founded at Machilipatnam in 1923 by Bhogaraju Pattabhi Sitaramayya.

4. Regional Rural Banks were set up under the RRB Act, 1976.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: D. All four statements are correct.

PRACTICE

Match List-I (committee) with List-II (subject):

List-I (committee)	List-II (subject)
A. Narasimham Committee (1991)	1. Creation of the State Bank of India
B. Nachiket Mor Committee (2013)	2. Governance of bank boards
C. Gorwala Committee (AIRCS)	3. Financial sector reforms
D. P. J. Nayak Committee (2014)	4. Financial inclusion and payments banks

(1) A → 4 B → 3 C → 1 D → 2

(2) A → 3 B → 4 C → 1 D → 2

(3) A → 3 B → 4 C → 2 D → 1

(4) A → 3 B → 1 C → 4 D → 2

Answer: (2) Narasimham: reforms; Nachiket Mor: inclusion, payments banks; Gorwala (AIRCS): SBI; P. J. Nayak: bank boards.

PRACTICE

Which one of the following statements is NOT correct?

A. Payments banks can give loans to small farmers.

B. Regional Rural Banks are owned by the Centre, a sponsor bank and the State in a 50:35:15 ratio.

C. The SARFAESI Act lets banks seize secured assets without going to court.

D. NABARD was set up in 1982.

Answer: A. Payments banks can accept deposits (up to ₹2 lakh) and offer payment services but cannot lend.

PRACTICE

Assertion (A): After the bank mergers that took effect in April 2020, India had 12 public sector banks.
Reason (R): These mergers were carried out under the Insolvency and Bankruptcy Code, 2016.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: the mergers were government decisions to create stronger banks; the IBC deals with insolvent companies.

PRACTICE

A person has ₹7 lakh in a savings account and fixed deposits in one scheduled bank, which fails. Under DICGC deposit insurance, how much is the person sure to get back?

A. ₹1 lakh

B. ₹7 lakh

C. ₹2 lakh

D. ₹5 lakh

Answer: D. DICGC insures up to ₹5 lakh per depositor per bank, counting all accounts in that bank together.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the structure of banking with each type's features. 2. Write the history from the Presidency banks to nationalisation, Lead Bank and NABARD. 3. Write the reforms after 1991: Narasimham I and II, bad-loan laws, governance, mergers and inclusion.

SCIENCE · Health: disease burden, epidemics and pandemics in India

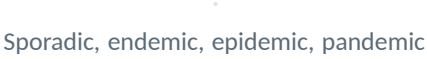
UNDERSTAND IT FIRST

What is it? Disease burden measures how much **illness, disability and early death** a disease causes in a population. It is counted in **DALYs (Disability-Adjusted Life Years)** = years lost to early death + years lived with disability. Diseases are **communicable** (spread by germs: TB, malaria, COVID-19) or **non-communicable (NCDs)** (not spread person to person: heart disease, diabetes, cancer). An **epidemic** is a sudden rise in cases; a **pandemic** is an epidemic that spreads across the world. Learn it as **how burden is**

measured → India's health transition → major communicable diseases → successes → NCDs and nutrition → epidemics and pandemics → laws and preparedness.

1. India's health transition

- India has gone through an **epidemiological transition**: in **1990**, communicable, maternal, neonatal and nutritional diseases caused most of the burden; today **NCDs** cause **about two-thirds of deaths** (**ICMR–PHFI India State-Level Disease Burden Initiative**). Leading causes now: **heart disease (ischaemic heart disease), COPD (lung disease), stroke, diabetes**, with **air pollution, high blood pressure, poor diet, tobacco and high blood sugar** as main risk factors. India still carries a **'double burden'**: old infections plus new lifestyle diseases. States in **earlier transition** (e.g. Bihar, UP) still have more infections; **Kerala, Tamil Nadu, Goa** have more NCDs.



Sporadic, endemic, epidemic, pandemic

2. Major communicable diseases

Disease	Cause and spread	India's response
Tuberculosis (TB)	Bacterium **Mycobacterium tuberculosis**; through the **air** (coughing); India has about **a quarter of the world's cases**.	**National TB Elimination Programme**; goal to end TB ahead of the SDG year **2030**; **Ni-kshay Poshan Yojana** (nutrition money); **100-day TB Mukt Bharat Abhiyaan (Dec 2024)**; free drugs; **BCG** vaccine.
Malaria	Parasite **Plasmodium** (falciparum, vivax); spread by the **female Anopheles** mosquito; common in **tribal, forest areas** (Odisha, Chhattisgarh, Jharkhand, NE, AP Agency).	**National Framework for Malaria Elimination**: zero cases by **2027**, eliminate by **2030**; bed nets, spraying; India has left the WHO's **'High Burden to High Impact'** group (2024).
Dengue, chikungunya	**Viruses**; spread by **Aedes aegypti**, which bites in the **daytime** and breeds in **clean stored water** (coolers, tyres, pots); dengue can cause **low platelets** and shock.	Under the **National Centre for Vector Borne Diseases Control (NCVBDC)**; no specific drug; **source reduction**; **16 May National Dengue Day**.
Filariasis, kala-azar, JE	**Lymphatic filariasis** (elephantiasis): worm spread by **Culex**; **kala-azar** (visceral leishmaniasis): parasite spread by the **sandfly** (Bihar, Jharkhand); **Japanese encephalitis**: virus spread by **Culex**, pigs and birds as hosts.	**Mass Drug Administration** for filariasis; kala-azar brought **below 1 case per 10,000** in every block (**2023**); **JE vaccine** in routine immunisation.
HIV/AIDS, leprosy, Nipah	**HIV**: virus; sex, blood, mother to child. **Leprosy**: bacterium **Mycobacterium leprae**. **Nipah**: a **zoonotic** virus from **fruit bats**; Kerala outbreaks since **2018**.	**NACP (1992)**, **NACO**; **HIV/AIDS Act 2017** (no discrimination); free ART; leprosy **eliminated as a public health problem (2005)** at

		national level, but new cases continue; Nipah: isolation, contact tracing.

- **Successes: smallpox** eradicated (India's last case **1975**; certified **1977**; worldwide **1980**); **guinea worm** (2000); **polio**: last case **13 January 2011** (Howrah), WHO certified India **polio-free on 27 March 2014**; **maternal and neonatal tetanus** eliminated (**2015**); **yaws** (WHO, **2016**); **trachoma** eliminated as a public health problem (WHO, **October 2024**). **Universal Immunisation Programme (1985)**, **Mission Indradhanush (2014)**, the **U-WIN** digital vaccine registry. **Regional issue in AP: chronic kidney disease of unknown cause** in **Uddanam** (Srikakulam district) has drawn special study and a dialysis and research centre at **Palasa**.
- **NCDs and nutrition:** the **National Programme for Prevention and Control of NCDs (NP-NCD**, earlier NPCDCS, **2010**) screens everyone over **30** for **hypertension, diabetes and common cancers** at **Ayushman Arogya Mandirs**; the **'75/25' initiative (2023)** aimed to put **75 million** people with hypertension or diabetes on standard care by **2025**. **NFHS-5 (2019–21):** children under 5: **stunted 35.5%, wasted 19.3%, underweight 32.1%**; **anaemia** in **57%** of women aged 15–49; **Poshan Abhiyaan (2018)** and **Anaemia Mukt Bharat** respond to this.

3. Epidemics and pandemics

- **Past pandemics in India: cholera** pandemics in the 19th century spread from the **Bengal delta**; **bubonic plague** from **Bombay (1896)** led to the **Epidemic Diseases Act, 1897**; the **1918 influenza ('Spanish flu')** killed an estimated **12–20 million** Indians, the worst hit of any country; **H1N1 'swine flu' (2009)**; the **1994 Surat plague** outbreak.
- **COVID-19:** caused by the virus **SARS-CoV-2**; India's first case on **30 January 2020** (Thrissur, Kerala), the same day the WHO declared a **Public Health Emergency of International Concern (PHEIC)**; the WHO called it a **pandemic on 11 March 2020**; national **lockdown from 25 March 2020** under the **Disaster Management Act, 2005**; the **Epidemic Diseases Act was amended (2020)** to protect health workers; vaccination began **16 January 2021** (**Covishield, Covaxin**), over **220 crore doses** through **CoWIN**; the WHO ended the emergency on **5 May 2023**.
- **Preparedness: Integrated Disease Surveillance Programme (IDSP, 2004)** and its digital **IHIP**; **National Centre for Disease Control (NCDC)**; **ICMR–National Institute of Virology, Pune**; **PM Ayushman Bharat Health Infrastructure Mission (PM-ABHIM, 2021)** for labs and critical-care blocks; the **'One Health'** approach links **human, animal and environmental health** (most new diseases are **zoonotic**). Globally, the **WHO Pandemic Agreement** was adopted by the World Health Assembly in **May 2025** (check its later progress).

EXAM TRAP

DALY = years lost to early death + years with disability; **transition**: NCDs **~2/3 of deaths** (ICMR–PHFI); top: **heart disease, COPD, stroke, diabetes**; risks: **air pollution, BP, diet, tobacco, sugar**; **double burden**; **TB** (M. tuberculosis; air; **~1/4 of world**; NTEP; **Ni-kshay Poshan**; **100-day campaign Dec 2024**; BCG); **malaria** (Plasmodium; **female Anopheles**; zero cases **2027**, eliminate **2030**; out of HBHI 2024); **dengue, chikungunya** (virus; **Aedes, daytime, clean water**; platelets; **16 May**); **filariasis (Culex; MDA)**; **kala-azar (sandfly; <1/10,000 in 2023)**; **JE (Culex; pigs)**; **HIV (NACP 1992, Act 2017)**; **leprosy (M. leprae; 2005)**;

Nipah (fruit bats; Kerala 2018); smallpox (1977/1980), guinea worm 2000, polio (last 13 Jan 2011 Howrah; certified 27 Mar 2014), MNT 2015, yaws 2016, trachoma Oct 2024; UIP 1985, Mission Indradhanush 2014, U-WIN; Uddanam CKDu (Palasa centre); NP-NCD (2010), screening 30+, 75/25; NFHS-5: stunting 35.5, wasting 19.3, underweight 32.1, anaemia women 57%; sporadic, endemic, epidemic, pandemic; **cholera (Bengal), plague 1896 → Epidemic Diseases Act 1897, 1918 flu 12–20 million**, H1N1 2009, Surat 1994; **COVID: first case 30 Jan 2020 (Thrissur) = PHEIC day; pandemic 11 Mar 2020; lockdown 25 Mar (DM Act 2005); 1897 Act amended 2020; vaccines 16 Jan 2021; 220 cr+ doses; emergency ended 5 May 2023; IDSP 2004, IHIP, NCDC, NIV Pune, PM-ABHIM 2021, One Health; WHO Pandemic Agreement May 2025**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. India was certified polio-free by the WHO in 2014.

2. Smallpox has been eradicated worldwide.

3. Dengue is spread by the Anopheles mosquito.

4. Non-communicable diseases now cause the majority of deaths in India.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 4 only

D. 1, 2, 3 and 4

Answer: C. Dengue is spread by the Aedes mosquito; Anopheles spreads malaria.

PRACTICE

Match List-I (disease) with List-II (vector):

List-I (disease)	List-II (vector)
A. Malaria	1. Sandfly
B. Dengue	2. Culex mosquito
C. Kala-azar	3. Female Anopheles mosquito
D. Lymphatic filariasis	4. Aedes mosquito

(1) A → 3 | B → 4 | C → 1 | D → 2

(2) A → 4 B → 3 C → 1 D → 2

(3) A → 3 B → 4 C → 2 D → 1

(4) A → 3 B → 1 C → 4 D → 2

Answer: (1) Malaria: Anopheles; dengue: Aedes; kala-azar: sandfly; filariasis: Culex.

PRACTICE

Which one of the following statements is NOT correct?

A. India's first COVID-19 case was reported in Kerala in January 2020.

B. The WHO declared COVID-19 a pandemic on 30 January 2020.

C. The national lockdown of March 2020 was imposed under the Disaster Management Act, 2005.

D. COVID-19 vaccination in India began on 16 January 2021.

Answer: B. On 30 January 2020 the WHO declared a Public Health Emergency of International Concern; it called COVID-19 a pandemic on 11 March 2020.

PRACTICE

Assertion (A): Antibiotics are the main treatment that cures dengue.
Reason (R): Dengue is caused by a virus.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: D. A is false: antibiotics act on bacteria, not viruses; dengue is treated with fluids, rest and monitoring of platelets. R is true.

PRACTICE

In some tribal forest belts, malaria cases occur at a fairly steady level year after year. How is the disease described in these areas?

A. Endemic

B. Epidemic

C. Pandemic

D. Sporadic

Answer: A. A disease constantly present at a steady level in an area is endemic; a sudden rise above that level would be an epidemic.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write DALY, the health transition and the double burden. 2. Write each communicable disease with cause, vector and programme, and the diseases eradicated or eliminated with years. 3. Write NCDs and NFHS-5 figures, the four levels of spread, historic pandemics, COVID-19 dates and the preparedness systems.

NOTEBOOK PAGE · Day 35: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Andhra Pradesh: land, climate, rivers

Area	Write
Land	Location, area, coastline, districts; three physical divisions; Eastern Ghats hill ranges north to south; highest peak; caves, gorge, lakes, islands
Climate and rivers	Four seasons; SW vs NE monsoon by region; wettest and driest areas; cyclones with years; rivers north to south with tributaries, dams, barrages, deltas; projects (Polavaram, Pattiseema, Handri-Neeva, Telugu Ganga, Galeru-Nagari)

ECONOMY: Indian banking

- Structure (scheduled/non-scheduled; commercial, RRB, SFB, payments, co-operative); history (presidency banks, PNB, Andhra Bank, SBI 1955, nationalisation 1969/1980, Lead Bank); reforms (Narasimham I and II, private licences, NPAs, SARFAESI, IBC, mergers, Jan Dhan, DICGC, 2025 amendments).

SCIENCE: Disease burden, epidemics and pandemics

Write: DALY and the health transition; communicable diseases with cause, vector and programme (TB, malaria, dengue, filariasis, kala-azar, HIV, leprosy, Nipah); diseases eradicated/eliminated with years; NCDs and nutrition (NFHS-5); sporadic/endemic/epidemic/pandemic; historic pandemics; COVID-19 dates; laws and systems (1897 Act, DM Act 2005, IDSP, PM-ABHIM, One Health, WHO Pandemic Agreement).